

Homework 2

You're only allowed to use the standard built-in Python 3 libraries with the exception of Numpy and Matplotlib.

Question 1. Plot the following trigonometric functions for $-5\pi \leq x \leq 5\pi$.

$$f(x) = x \sin^2(x), \quad g(x) = -x \sin^2(x)$$

- The title of the plot should read "Trigonometric Functions."
- The x -axis label of the plot should read " x -axis."
- The y -axis label of the plot should read " y -axis."
- The legends should appropriately denote the different functions by their definition.
- The graphs should have smooth continuous curves.

Question 2. Download and save the novel "[The Cosmic Computer](#)" by Piper. Read the saved text file into Python and plot a bar graph of the character frequency (case insensitive).

- The title of the plot should read 'Character Frequency in "The Cosmic Computer" by Piper.'
- The x -axis label of the plot should read "Alphabet."
- The y -axis label of the plot should read "Frequency."
- The bars should be sorted in ascending order.

Question 3. Using `matplotlib.pyplot.imshow` and an eight by eight binary Numpy array, plot the chess board pattern.

- The x -ticks should be letters from A to H, A being at the left.
- The y -ticks should be numbers from 1 to 8, one being at the bottom.
- The title should be "Chess Board Pattern."

Question 4. Following are the two matrices from the last homework,

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 4 & 2 & 2 \end{bmatrix}$$

Use Numpy to multiply them and print the product $\mathbf{AB}$. Does it match with the one you carried out by hand?

Question 5. The Gregory series is given as,

$$f(n) = \sum_{i=1}^{i=n} \frac{4(-1)^{i+1}}{2i-1} = \frac{4}{1} - \frac{4}{3} + \frac{4}{5} - \cdots + \frac{4(-1)^{n+1}}{2n-1}$$

- 1) Without using any loops, i. e., using vectorised operations over Numpy arrays, plot the following values,

$$\{(n, f(n)) : 1 \leq n \leq 100\}$$

- 2) Plot the following error curve,

$$\{(n, (\pi - f(n))^2) : 1 \leq n \leq 100\}$$

Label the axis and title each plot appropriately. What is $\lim_{n \rightarrow \infty} f(n)$?

[somewhere between 3 and 3.2](#)

Question 6. Consider the following study results in table 1,

Point	Real	Predicted
1	−	+
2	−	−
3	−	−
4	−	+
5	+	−
6	+	+
7	+	+
8	+	+

TABLE 1. Real and Predicted Labels

State all the points that are,

- 1) True Negatives
- 2) True Positives
- 3) False Positives
- 4) False Negatives

- 1) 2, 3
- 2) 6,7,8
- 3) 1,4
- 4) 5

Question 7. Give the confusion matrix for table 1.

	Predicted −	Predicted +
Real −	TN	FP
Real +	FN	TP

TABLE 2. Binary Confusion Matrix

	Predicted −	Predicted +
Real −	2	2
Real +	1	3

Table 3. Confusion Matrix

Question 8. Again, for table 1, give the following in exact fractions.

- 1) Baseline Accuracy
- 2) Accuracy

$$\frac{\text{Total Correctly Classified}}{\text{Total Examples}}$$

- 3) Precision

$$\frac{\text{True Positives}}{\text{Total Examples Classified as Positive}}$$

- 4) Recall

$$\frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

5) F1 Score

$$\frac{2TP}{2TP + FP + FN}$$

1) 5 / 8

2) 3 / 5

3) 3 / 4

4) $2(3)/2(3) + 2 + 1 = 6/9 = 2/3$

Question 9. Often (but not always) how are precision and recall related?

Precision is how overall correct you are and recall is how many true positives you catch. If you focus on recall, you catch more true positives but also end up with more false positives which is less precise but may be better depending on what your predicting.

Question 10. Between precision and recall, which is a better metric for,

1) Predicting good restaurants.

2) Predicting cancer.

precision is better for rating non important things so that you get accurate info on something like restaurant ratings. For something important like cancer, its much more important to get as many positives as possible even if it means some may be false, its better to be safe than sorry.

Question 11. Between precision and recall, which metric is used in the Receiver Operator's Curve (ROC). What does the area under this curve tell us about a prospective classifier?

More area under the curve means more true positives.

SUBMISSION INSTRUCTIONS

1) Submit a PDF that answers all the questions and includes any plots that the assignment asks for.

Circle and/emphasize your final answer wherever possible.

2) Submit a Python file, e. g., `plot.py` that generates all the plots one by one.

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